DSL-340 MkIII – Double Pass Particulate Monitor

Measuring in mg/m³ using DDP

Commissioning Guide with OI V1.1

This Installation Guide is intended for Engineers familiar with the full Installation and Operator Manuals including information and advice regarding choosing a suitable location, electrical safety and all warning messages as this is not covered in this guide.

Caution; care must be taken

All installations and connections must be made by suitably qualified and experienced electrical / mechanical engineers with proper regard to good practice and compliance with applicable legislation.

Caution; care must be taken

The instrument contains sensitive optics and electronics and must be protected from physical shock, vibration and exposure to moisture at all times. Always handle with care, replace lids and covers when not in use and protect from water ingress.

System Overview

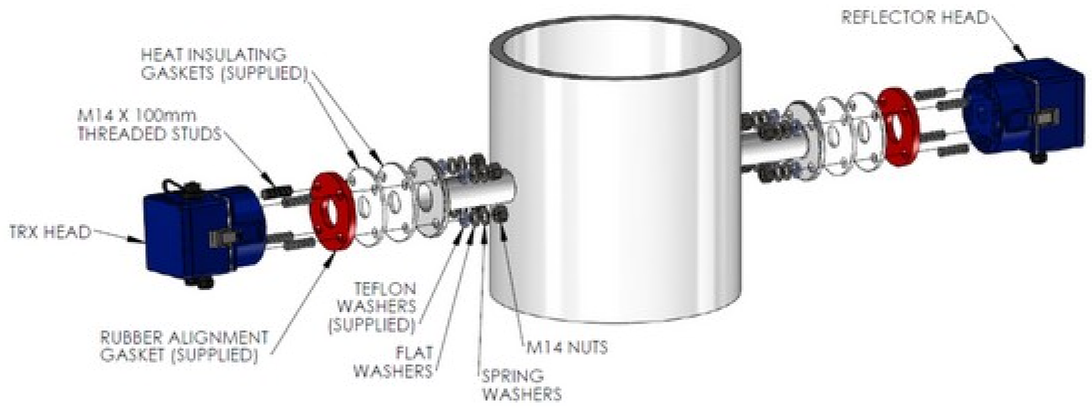

Figure 1 – System Overview

Set-up and Calibration

Set-up, calibration and commissioning involves performing the following steps, which must be completed in order.

1. **Perform a Head Alignment**
2. **Perform an Automatic Gain Correction**
3. **Define the Path length**
4. **Calibration of Particle Density**
5. **Configure the Outputs**
6. **Configure the Automatic Calibration Check (if required)**
7. **Save a Backup of the Settings**

Perform a Head Alignment

This optimises the mounting angle of the TRX and Reflector heads to achieve the largest possible measured signal.

The DSL-340 TRX produces a narrow optical beam that is approximately 3 degrees wide. To achieve a measurable signal this beam must overlap with the aperture of the Reflector head. This means that the angular alignment between the mounting flanges attached to the duct must be less than ±2 degrees, as defined in the installation manual.

To enable adjustment of the heads the DSL-340 Air-Purge Bodies must be fixed to the mounting flanges with the rubber flange gaskets (supplied) in between, as shown in Figure 2. The angle of the head can then be adjusted by tightening the fixing nuts and compressing the flange gasket in an uneven manner.

Please note that the nuts used to compress the flange gasket should not be over tightened or too loose. The recommended torque range is 10 to 50Nm. The flange gasket provides only a small level of adjustment and cannot correct for angular errors between the flanges of more than the specified ±2 degrees.

The process of adjustment is to set all four nuts to the minimum torque (10Nm) then individually tighten each one to compress the gasket and hence change the angle of the mounting. When the final alignment has been reached always check that all four nuts are sufficiently tight (at least 10Nm), since strongly adjusting one nut can result in others becoming loose.

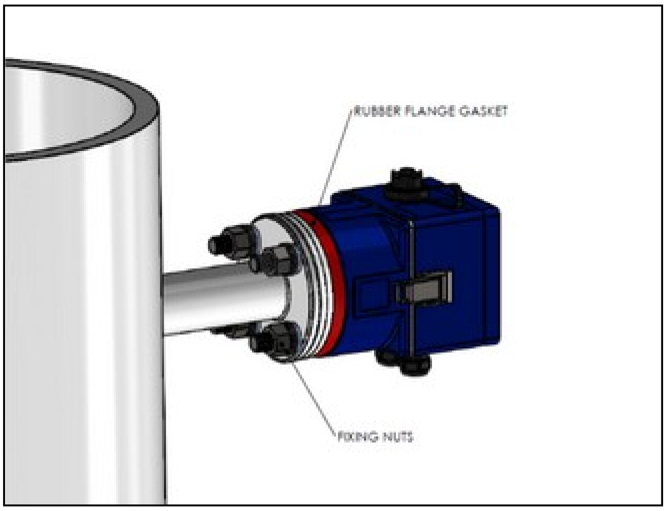

Figure 2: Air-Purge Body mounted to flange with the rubber flange gasket

The following steps outline the procedure for completing the alignment of the heads. This process assumes the use of a Laser Alignment Tool that is available as an optional extra from DynOptic Systems.

1. Attach the Laser Alignment Tool to the Reflector air-purge body and turn it on. Remove the back of the tool and view through the window of the instrument to see the red laser spot on the far side of the stack. The angle of the Reflector purge head can now be adjusted, using the mounting nuts, to approximately align the laser spot to the centre of the TRX flange on the far side of the stack.

   Remove the laser alignment tool and attach the Reflector head to its air-purge body.

2. Attach the Laser Alignment Tool to the TRX air-purge body on the other side of the stack. Repeat the alignment process to align the laser spot to the centre of the Reflector flange on the far side of the stack.

   Remove the laser alignment tool and attach the TRX head to its air-purge body.

The two heads are now well aligned but to get the optimum alignment it is necessary to monitor and maximise the measured signal, this is done by using the Utility Software.

Connecting a PC to the DSL-340

Warning: The Utility Software and the USB Driver MUST BE INSTALLED ON THE PC FIRST. See the Operators Manual for full instructions.

With the driver installed and com port assigned, connect your PC to the DSL-340 using a standard USB cable (type A to Type B).

The USB connection can be found on top of the TRX head (under a weather-proof cap) or inside the TRX enclosure itself (on the circuit board inside).

Power up the instrument.

NOTE: Error messages may be displayed during set-up, ignore these messages until after the set-up is completed.

Using the Utility software connect to the DSL-340 and select the ‘Settings’ tab, then select the Alignment Tool button. This will bring up the Alignment Tool window which allows the operator to monitor the measured signal strength as shown in Figure 3.

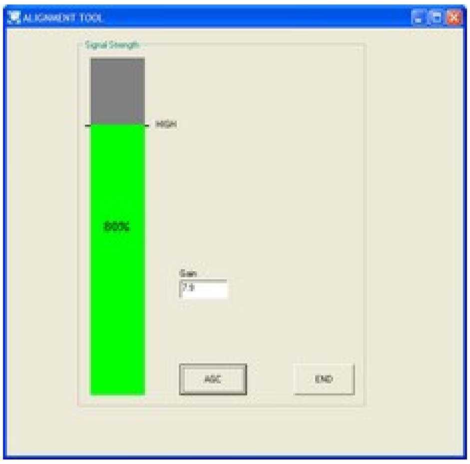

Figure 3 - Alignment Tool

Warning: The maximum signal can be more than 80%. Always perform an AGC and re-align for maximum signal.

The alignment tool window has a signal strength bar which increases in height with increases in the measured signal. If the measured signal goes too high (or too low) for the gain range currently set, the colour of the bar turns from green to red. When the bar turns red, click on the AGC button to adjust the instrument Gain to return the bar to green.

Please note: the measured signal is slightly damped so it takes a few seconds for the signal strength bar to respond to an adjustment in the head alignment.

1 Adjust the angle of the TRX head by using the mounting nuts to maximise the displayed signal level i.e. to maximise the height of the bar. If the bar goes too high and turns red, stop adjusting and click the AGC button to change the instruments Gain setting, then continue adjusting the angle of the head. Stop adjusting the angle of the head when the maximum signal strength has been achieved with minimum Gain. Ensure that all four mounting nuts are tight (at least 10Nm).

2 There is no need to adjust the angle of the Reflector head. The Reflector head uses corner cube reflectors whose performance is very insensitive to angle.

It is recommended that the instrument is left powered for at least a further 20minutes to stabilise and warm-up before proceeding with the next stage. Select ‘END’ to exit the Alignment Assistance Tool.

Perform an Automatic Gain Correction

An AGC alters the instruments gain value to achieve the optimum signal strength of 4V. It is essential that the ~4.0V is achieved under clear path conditions. Clear path conditions are defined as follows:

1. The instrument must be properly and permanently installed as per the separate Installation Manual and the heads must be properly aligned as described above.
2. The instrument must have been powered up and left to settle for a minimum of 1 hour.
3. The optical surfaces of the instrument must be perfectly clean and free from dirt, dust, moisture, or oil.
4. The optical path between the TRX and the Reflector must be clear and free from any obstruction.
5. The duct or stack must be free from any dust, smoke, soot, steam, or particulate (i.e. the process behind the stack must be shut down).

Once clear path conditions have been achieved the gain correction should be performed by clicking the AGC button on the Settings tab in the Utility Software.

This will cause the instrument to automatically calculate and store a new gain value to achieve the optimum signal strength ~4.0V.

Note that the DSL-340 should not be operated with a gain of more than 2000. The instrument is capable of setting higher gains, but these are for use in alignment and setup only. If the AGC produces a gain higher than 2000 then either the path length being used is too long for the particular instrument or the head optics are dirty or they have not been aligned correctly. If any of these two conditions are not correct then there may be an instrument fault which must be reported to a DynOptic trained engineer or your local distributor.

The achieved signal strength can be seen in the Measure Signal box, alongside the Gain box, in the Utility software. See Figure 4.

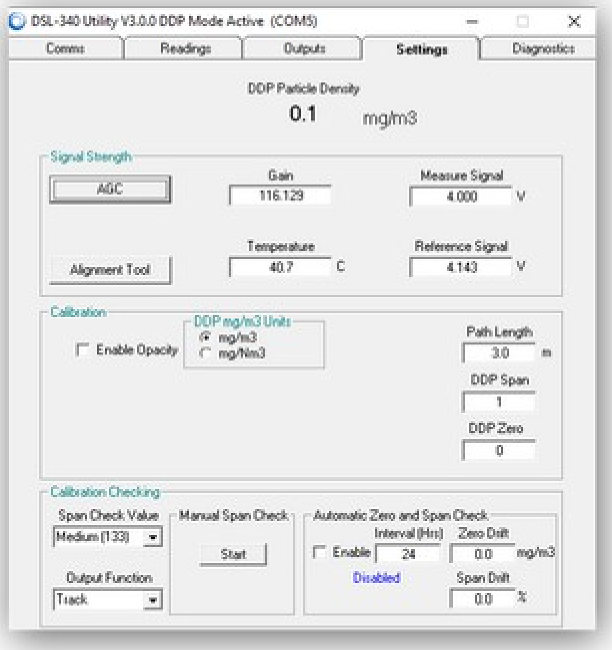

Figure 4 – Settings Tab

Define the Path length

Path length is defined as the distance in metres between the face of the flange on the Air-Purge body of the TRX, and the same point on the Air-Purge body of the Reflector. It is necessary to enter this value to accuracy of ±0.05m (5cm) on the Settings tab.

Calibration of Particle Density

The relationship between the measured DDP signal and Particle Density is approximately linear but the slope (DDP Span) and intercept (DDP Zero) will vary for each installation, based on the type, shape, size and colour of the particles in the gas stream.

The DDP Span and DDP Zero values cannot be pre-determined with factory settings and must instead be determined on site. The DDP Zero can be set by measuring a clear path. The DDP Span requires comparison of the instrument readings with an independent reference measurement. Making this comparison and adjusting the DDP Span value accordingly will complete the calibration for the instrument.

The nationally recommended standard reference measurement method should be used. This is likely to be a gravimetric analysis based on isokinetic sampling of the gas stream.

Please refer to the Operator Manual for full instructions.

Once the new DDP Span and DDP Zero values have been determined they can be entered on the Settings tab of the Utility.

Configure the Outputs

Once the instrument is calibrated the 4-20mA and Level Alarm outputs can be configured as required for integration with the wider plant systems. On the Utility program select the Outputs tab.

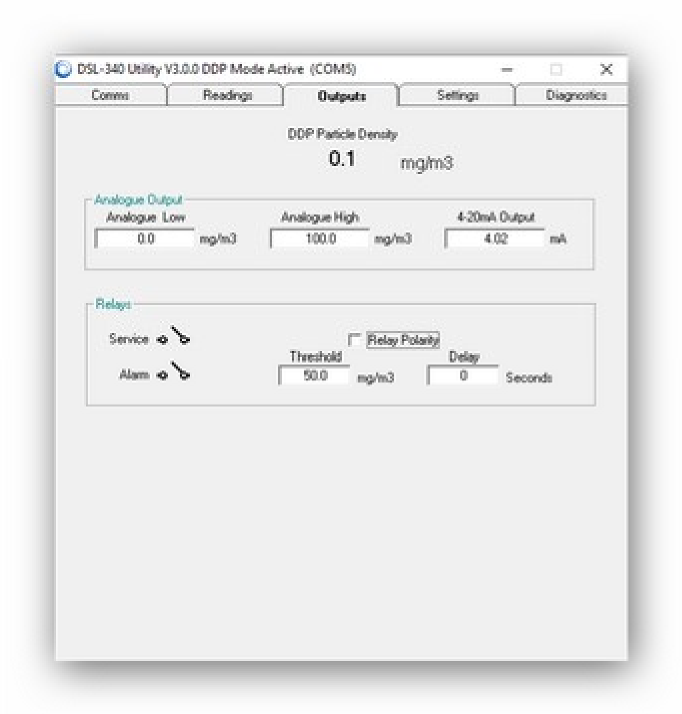

Figure 5 – Outputs Tab

Analogue Low and Analogue High

These two parameters set the upper and lower scaling points of the 4-20mA analogue output.

Set the Analogue Low value to the reading at which you want the analogue output to produce 4mA.

Set the Analogue High value to the reading at which you want the analogue output to produce 20mA.

4-20mA Output

Note; the mA box is a read only parameter and shows a calculated indication of the expected 4-20mA output for each analogue output channel - taking into account the An HI/LO values. The values shown here are calculated values, they are not measured values from the actual 4-20mA outputs.

Relays

This section shows the current status of the two (2) relays. The graphical image associated with each relay indicates whether the relay contact is currently open or closed. Contacts are open in a de-energised state (i.e. N/O).

The first relay is a service indicator that will go into an alarm condition in the event that the instruments' self-checking identifies a fault.

The second relay is the measurement reading alarm. This operates when the measurement reading exceeds a set Threshold value. The operation of this alarm can be delayed using the Delay setting.

Threshold

This parameter defines the measurement reading at which the alarm relay will trip.

Delay

This value determines the continuous length of time (in seconds) for which the measurement reading must have exceeded the associated threshold value, before the alarm will activate and the relay will change state. Delaying the activation of the relay in this way can prevent borderline level changes from "dithering" the relay state. Only genuine, sustained readings in excess of the threshold will actually trigger the alarm.

Relay Polarity checkbox

Checking this box changes the default state of the relays.

With the checkbox UNCHECKED, the associated relay will be de-energised in a below threshold condition and the relay status indicator will show an open contact in the same below threshold condition. In this polarity the relay will be energised when the reading exceeds the threshold.

With the checkbox CHECKED, the relays will be energised in a below threshold condition, and the relay status indicator will show a closed contact in the same below threshold condition. In this polarity the relay will de-energise when the reading exceeds the threshold. This condition represents the recommended failsafe operation.

Configure the Automatic Calibration Check

The instrument contains a built in electronic calibration check procedure which can be used to confirm that the instrument is still operating within its design parameters. During an automatic calibration check the instrument first performs a zero check and then a span check. The whole procedure takes about 95 seconds during which the data invalid flag will be enabled.

The automatic calibration check can be configured using the setting tab of the Utility. First select the required "Span Check Value" from the drop down menu then select the "Output Function" (Track or Hold). Finally enter the time interval in hours (default is 24). Selecting the "Enable" tick box will start the automatic calibration checking routine.

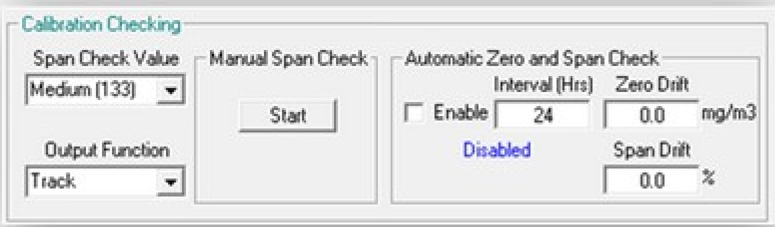

Figure 6 – Settings Tab: Calibration Checking

Please refer to the Operator Manual for full instructions.

Save a Backup of the Settings

On the "Diagnostics" tab of the Utility click on the "Backup to PC" button. This will save the current instrument settings to a backup file in the location: 'C\DynOptic\DSL-340 Utility\BACKUP' on the PC. The filename of the backup file will incorporate the date, time and electronic serial number of the instrument at time of backup. It is highly recommended that you use this button to take a backup of your instrument settings after successfully completing your installation, calibration and commissioning. The backup file can be used to restore the instruments settings, should the live settings ever be lost or become corrupted.

Error Messages

If any error messages are displayed on the Readings tab, click on the message and it should clear. If the message(s) re-appear, check the fault code list below, or see the 'Troubleshooting Guide' in the Operator Manual for a full description and possible solutions.

Fault Code	**Description**
LO REF-VIS	Indicates that the Reference Signal has fallen below 1.5V. This error is usually caused by a dim (or failed) LED in the TRX head.
HI REF-VIS	Indicates that the Reference Signal has risen above 4.9V. The DSL-330 electronics can only measure a maximum signal of 5.0V.
HI MEAS-VIS	Indicates that the Measure Signal has risen above 4.9V. The DSL-330 electronics can only measure a maximum Measure Signal of 5.0V.
LO MEAS-VIS	This message indicates that the optical transmission has fallen below 50% this indicates that the DDP signal is likely to be degraded. This fault is usually caused by dirty optics.
"HEAd" (only displayed on the OI)	Indicates that the OI has lost communications with the DSL-340 TRX Head. This fault is usually caused by a problem with the cable between the OI and the TRX Head.
"CrC" (only displayed on the OI)	Indicates that the OI has regular data corruption occurring in the communications between the OI and the TRX Head. Corruption is rare and is usually caused by RF interference, electromagnetic radiation, or a poor connection in the wiring.

Operation of the OI

The OI has a 4 digit LED display, four laminated push-buttons, and two LED alarm indicators.

The **Cycle** button cycles you through the available parameters on the Page. It can also be used to cancel data entry.

The **Next** button initiates data entry and moves between digits during data entry.

The **Increase** button is used to increment a digit or move the decimal point during data entry.

The **Enter** button is used to end data entry and store the altered value. It can also be used to re-display a parameters “name”.

The level alarm LED will illuminate when the threshold of the level alarm is exceeded.

The service alarm LED will illuminate when the instrument detects a fault condition that requires attention.

The display will display just one parameter at a time, but the Cycle button can be used to cycle the display through all the available parameters in turn. When cycling through parameters the display will initially show the “name” of the parameter.

Once cycling is ended, and the display has settled on one parameter for more than one second, the display will switch to show the “value” of that parameter.

The value of a parameter can be altered by initiating data entry mode, which is achieved by pressing the Next button. On initiating data entry mode the first digit of the value will begin to repeatedly flash indicating that the value of that digit can now be incremented. Only one digit can be increased in value at any one time, so select a digit by moving between them with the Next button. Increase the value of a selected digit using the increase button.

It is possible to exit data entry mode without saving an altered value (i.e. to abandon data entry) by pressing the Cycle button before pressing Enter during data entry.

Moving between Pages

The parameters available in the display are arranged on two “pages”.

Page 0 is made up of parameters which are more commonly used, such as output settings, damping, and alarm delays.

Page 1 comprises parameters which are less commonly used or that can be used to alter the calibration or fundamental setup of the instrument.

Switching between Pages is achieved by selecting the Page parameter and toggling between 0 and 1. Because Page 1 contains parameters which can alter the calibration and fundamental setup of the instrument, it is password protected, so upon choosing Page 1 the message PAS? will show briefly, and the user will then be prompted to enter a 4 digit number (the password) using the usual data entry method. *The default password is 0000.*

If the correct number is entered, the message yES will appear and the display will move to Page 1. If the wrong password is entered, the message no will appear and the display will default back to Page 0.

For a full guide to the operation of the OI see the Operator Manual.

Revision Control

Version	Revision Date	Revision Details	Author
V1.0	11/02/2014	Original (as per V1.3 of MkII manual).	Dominic Sheedy
V1.1	09/02/2106	Updated for new DSL-3xx UTILITY (V3.0.0)	Colin Edge

All technical details and specifications are subject to change without notice